Project Title: Password Strength Checker

Course: Build Your Own Project – VITyarthi

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Introduction

Intoday's digital world, passwords act as the primary line of defense for protecting user data and digital identities. However, weak or predictable passwords remain a major security risk, as attackers often exploit common patterns, simple structures, or dictionary words to gain unauthorized access.

This project presents a Password Strength Checker that evaluates the security level of a password based on several parameters such as length, character diversity, entropy, and common-password detection. The goal is to educate users on choosing safer passwords and to provide real-time feedback on improving password quality.

Problem Statement

Many users choose weak passwords that can be easily guessed or cracked using brute force, dictionary attacks, or social engineering.

There is a need for a simple yet effective system that can:

*Detect weak passwords

*Suggest improvements

*Help users build stronger, more secure passwords

This project aims to solve this issue by developing a Password Strength Checker using Python.

Functional Requirements

1. The system must accept a password input from the user.

2. It must evaluate strength based on:

*Length of password

*Uppercase and lowercase characters

*Numerical digits

*Special characters

*Presence in common-password database

*Entropy score

3. The system must categorize the password as:

*Weak

*Medium

*Strong

*Very Strong

4. The system must display the reason behind the strength level (optional enhancement).

5. Non-Functional Requirements

1. Performance: Password evaluation must occur in less than 1 second.

2. Security: Passwords should not be saved, logged, or transmitted.

3. Usability: Simple, clean interface that anyone can use.

4. Reliability: Must provide consistent results for all inputs.

5. Maintainability: Code should be modular and easy to update.

System Architecture

Architecture Flow:

User Input → Input Module → Password Evaluation Engine

→ Entropy Calculator → Strength Decision Module → Output Result

Modules Used:

Input Handler

Pattern Checker

Entropy Calculator

Common Password Validator

Output Formatter

Design Diagrams

(Insert the diagrams after I generate them — tell me “Generate diagrams” and I will provide images.)

You must include the following:

1. Use Case Diagram

```
+-------------+
|    User     |
+-------------+
      |
      V
+-----------------------+
| Password Checker App |
+-----------------------+
 /      |         \
Check Length  |     Check Complexity
     Check Patterns
         Check Dictionary
            Calculate Entropy
```

2. Workflow Diagram

```
START
  ↓
Input password
  ↓
Check length → Check character types → Check dictionary words → Check patterns → Calculate entropy
  ↓
Generate score
  ↓
Display rating + suggestions
  ↓
END
```

3. Sequence Diagram

```
User → System: Enter password
System → Length Module: Validate length
System → Complexity Module: Analyze characters
System → Dictionary Module: Check common words
System → Pattern Module: Detect sequences
System → Entropy Module: Calculate entropy
System → User: Provide rating + suggestions
```

4. Class / Component Diagram

```
+---------------------+
|  PasswordChecker    |
+---------------------+
| +check_length()     |
| +check_complexity() |
| +check_dictionary() |
| +check_patterns()   |
| +calculate_entropy()|
| +generate_score()   |
+---------------------+
```

5. ER Diagram – (Not required since no databasestorage is used. Write “Not Applicable”)

```
USER_TABLE
------------
User_ID (PK)
Password
Strength_Score
Timestamp
```

Design Decisions & Rationale

1. Python was chosen because it offers:

*Fast development

*Built-in regex libraries

*Easy entropy calculation

2. Entropy-based evaluation gives more accurate strength measurement than simple rule checking.

3. No database used to ensure user privacy.

4. Modular design ensures easy future upgrades (GUI, Mobile App, Website version etc.).

Implementation Details

Technologies / Tools Used

Python (Core Logic)

Regular Expressions

Math module (Entropy calculation)

Command-line interface

Code Structure

```
password_checker/
```

```
│
├── password_strength_checker.py
├── common_passwords.py (optional)
└── README.md
```

Main Logic Components

1. Pattern Checking – Detects uppercase, lowercase, digits, symbols

2. Entropy Calculation – Uses mathematical formula

3. Common Password Detection – List contains commonly used passwords

4. Output Module – Displays strength level

Python Code Used (Full)

```
import re
import math

common_passwords = ["password", "123456", "qwerty", "admin", "letmein"]

def calculate_entropy(password):
    pool = 0
    if re.search(r"[a-z]", password): pool += 26
    if re.search(r"[A-Z]", password): pool += 26
```

```
    if re.search(r"[0-9]", password): pool += 10
    if re.search(r"[^A-Za-z0-9]", password): pool += 32
    return len(password) * math.log2(pool) if pool else 0

defcheck_strength(password):
    score = 0

    iflen(password) >= 8: score += 1
    ifre.search(r"[A-Z]", password): score += 1
    ifre.search(r"[a-z]", password): score += 1
    ifre.search(r"[0-9]", password): score += 1
    ifre.search(r"[^A-Za-z0-9]", password): score += 1
    ifpassword.lower() not in common_passwords: score += 1

    entropy = calculate_entropy(password)

    ifentropy < 28:
        return "Weak"
    elifentropy < 36:
        return "Medium"
    elifentropy < 60:
        return "Strong"
    else:
        return "Very Strong"

password = input("Enter a password: ")
print("Password Strength:", check_strength(password))
```

```python
import re
import math

common_passwords = ["password", "123456", "qwerty", "admin", "letmein"]

def calculate_entropy(password):
    pool = 0
    if re.search(r"[a-z]", password): pool += 26
    if re.search(r"[A-Z]", password): pool += 26
    if re.search(r"[0-9]", password): pool += 10
    if re.search(r"[^A-Za-z0-9]", password): pool += 32
    return len(password) * math.log2(pool) if pool else 0

def check_strength(password):
    score = 0

    if len(password) >= 8: score += 1
    if re.search(r"[A-Z]", password): score += 1
    if re.search(r"[a-z]", password): score += 1
    if re.search(r"[0-9]", password): score += 1
    if re.search(r"[^A-Za-z0-9]", password): score += 1
    if password.lower() not in common_passwords: score += 1

    entropy = calculate_entropy(password)

    if entropy < 28:
        return "Weak"
    elif entropy < 36:
        return "Medium"
    elif entropy < 60:
        return "Strong"
    else:
        return "Very Strong"

password = input("Enter a password: ")
print("Password Strength:", check_strength(password))
```

Screenshots / Results

Weak password result

```python
import math

common_passwords = ["password", "123456", "qwerty", "admin", "letmein"]

def calculate_entropy(password):
    pool = 0
    if re.search(r"[a-z]", password): pool += 26
    if re.search(r"[A-Z]", password): pool += 26
    if re.search(r"[0-9]", password): pool += 10
    if re.search(r"[^A-Za-z0-9]", password): pool += 32
    return len(password) * math.log2(pool) if pool else 0

def check_strength(password):
    score = 0

    if len(password) >= 8: score += 1
    if re.search(r"[A-Z]", password): score += 1
    if re.search(r"[a-z]", password): score += 1
    if re.search(r"[0-9]", password): score += 1
    if re.search(r"[^A-Za-z0-9]", password): score += 1
    if password.lower() not in common_passwords: score += 1

    entropy = calculate_entropy(password)

    if entropy < 28:
        return "Weak"
    elif entropy < 36:
        return "Medium"
    elif entropy < 60:
        return "Strong"
    else:
        return "Very Strong"

password = input("Enter a password: ")
print("Password Strength:", check_strength(password))
```

```
Enter a password:  123456
Password Strength: Weak
```

Medium result

```python
common_passwords = ["password", "123456", "qwerty", "admin", "letmein"]

def calculate_entropy(password):
    pool = 0
    if re.search(r"[a-z]", password): pool += 26
    if re.search(r"[A-Z]", password): pool += 26
    if re.search(r"[0-9]", password): pool += 10
    if re.search(r"[^A-Za-z0-9]", password): pool += 32
    return len(password) * math.log2(pool) if pool else 0

def check_strength(password):
    score = 0

    if len(password) >= 8: score += 1
    if re.search(r"[A-Z]", password): score += 1
    if re.search(r"[a-z]", password): score += 1
    if re.search(r"[0-9]", password): score += 1
    if re.search(r"[^A-Za-z0-9]", password): score += 1
    if password.lower() not in common_passwords: score += 1

    entropy = calculate_entropy(password)

    if entropy < 28:
        return "Weak"
    elif entropy < 36:
        return "Medium"
    elif entropy < 60:
        return "Strong"
    else:
        return "Very Strong"

password = input("Enter a password: ")
print("Password Strength:", check_strength(password))
```

```
Enter a password:  pokemon
Password Strength: Medium
```

Strong/Very Strong result

```python
import math

common_passwords = ["password", "123456", "qwerty", "admin", "letmein"]

def calculate_entropy(password):
    pool = 0
    if re.search(r"[a-z]", password): pool += 26
    if re.search(r"[A-Z]", password): pool += 26
    if re.search(r"[0-9]", password): pool += 10
    if re.search(r"[^A-Za-z0-9]", password): pool += 32
    return len(password) * math.log2(pool) if pool else 0

def check_strength(password):
    score = 0

    if len(password) >= 8: score += 1
    if re.search(r"[A-Z]", password): score += 1
    if re.search(r"[a-z]", password): score += 1
    if re.search(r"[0-9]", password): score += 1
    if re.search(r"[^A-Za-z0-9]", password): score += 1
    if password.lower() not in common_passwords: score += 1

    entropy = calculate_entropy(password)

    if entropy < 28:
        return "Weak"
    elif entropy < 36:
        return "Medium"
    elif entropy < 60:
        return "Strong"
    else:
        return "Very Strong"

password = input("Enter a password: ")
print("Password Strength:", check_strength(password))
```

```
Enter a password:  MyS3cureP@ss
Password Strength: Very Strong
```

Testing Approach

Test Cases Used

1. Length testing: very short passwords
2. Character variety testing:

Only uppercase

Only numbers

Only symbols

3. Mixed input testing:

Password:

abc123

Abc!123

4. Common password testing:

Password:

123456

5. Complex password testing:

Password:

u!R2@fKbZ

Expected Outputs

Short/simple → Weak

Medium variety → Medium

Complex → Strong/Very Strong

Challenges Faced

1. Designing accurate entropy thresholds

2. Creating balanced scoring rules

3. Avoiding false positives for strong passwords

4. Handling edge-case inputs (unicode, spaces, empty password)

Learnings & Key Takeaways

*Learned importance of password security

*Understood entropy and randomness

*Practiced Python regular expressions

*Gained experience in modular code design

*Understood how to analyze user input safely

Future Enhancements

1. GUI Interface using Tkinter or PyQt

2. Web-based version using Flask

3. Mobile App using Kivy

4. Real-time suggestions to improve password

5. Integration with password managers

6. AI-based password strength prediction

References

1. Python Official Documentation – https://docs.python.org

2. NIST Password Guideline

3. OWASP Authentication Guide

4. Online research on entropy and password strength formulas